



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE (Advanced)

FULL SYLLABUS

27-04-2024

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE ADVANCED)

ANSWER KEY

PAPER-1 (OPTIONAL)

PART-1 : PHYSICS

SECTION-II (i)	Q.	1	2	3	4	5			
	A.	1.41	150.00	0.06	16.64	0.50			
SECTION-II (ii)	Q.	6	7	8	9	10			
	A.	248	3	5	1800	126			
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	8	3	3	3	5	4	2	6

PART-2 : CHEMISTRY

SECTION-II (i)	Q.	1	2	3	4	5			
	A.	7.00	4.00	24.63	45.15	15.05			
SECTION-II (ii)	Q.	6	7	8	9	10			
	A.	6	3	105	24	9			
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	7	6	6	2	4	3	9	8

PART-3 : MATHEMATICS

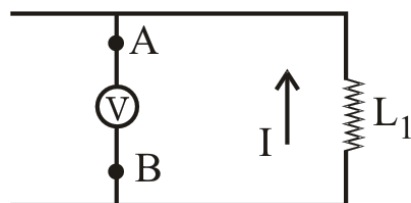
SECTION-II (i)	Q.	1	2	3	4	5			
	A.	0.50	0.40	0.25	2.00	2.00			
SECTION-II (ii)	Q.	6	7	8	9	10			
	A.	18	17	40	59	36			
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	1	1	1	2	1	1	2	1

HINT – SHEET

PART-1 : PHYSICS

SECTION-II (i)

1. Ans (1.41)

For very small ω circuit becomes

$$\therefore V_{AB} = I_{\text{rms}} Z = \frac{i_0}{\sqrt{2}} \omega L_1$$

2. Ans (150.00)

$$mg + \rho_g \times \frac{4}{3} \pi r^3 g = \rho_a \times \frac{4}{3} \pi r^3 g$$

$$\rho = \frac{PM}{RT} \Rightarrow \frac{\rho_g}{Mg} = \frac{\rho_a}{Ma}$$

$$P_a = \frac{10^5 \times 29 \times 10^{-3}}{\frac{25}{3} \times 300} = \frac{29}{25} \text{ kg/m}^3$$

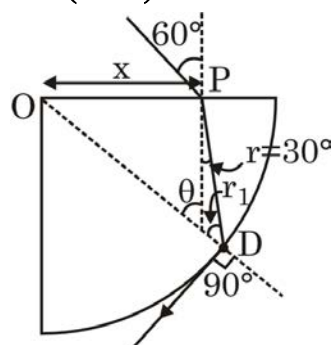
$$P_g = \frac{16}{25} \text{ kg/m}^3$$

$$\frac{4}{3} r^3 \times \frac{13}{25} = \frac{585\pi}{2} \times 10^{-3}$$

$$r^3 = \frac{125 \times 27}{8} \times 10^{-3}$$

$$r = 7.5 \times 10^{-1} \text{ m} = 75 \text{ cm}$$

3. Ans (0.06)



$$(1) \sin 90^\circ = \sqrt{3} \sin r_1$$

$$\sin r_1 = \frac{1}{\sqrt{3}}$$

$$(1) \sin 60^\circ = \sqrt{3} \sin r$$

$$\Rightarrow r = 30^\circ$$

$$\theta = 30 + r_1$$

Now in $\triangle OPD$

$$\frac{\sin r_1}{x} = \frac{\sin(90^\circ + 30^\circ)}{R} \Rightarrow \frac{1}{\sqrt{3}x} = \frac{\sqrt{3}}{2 \times 9} \Rightarrow x = 6\text{cm}$$

4. Ans (16.64)

$$d \sin \theta \pm (\mu - 1) t = \Delta x$$

$$d \left(\frac{3\lambda}{2d} \right) \pm \frac{3\lambda}{2} = \Delta x$$

$$\Delta x = 3\lambda$$

Hence maximum intensity will occur at 'O'.

$$I = 4I_0 = 16.64 \text{ W/m}^2$$

5. Ans (0.50)

$$a = \omega^2 A$$

$$f_1 = (1)\omega^2 A \leq 6 \quad \dots(1)$$

$$f_2 = 3\omega^2 A \leq 12$$

$$\omega^2 A \leq 4 \quad \dots(2)$$

$$\Rightarrow \omega^2 A = 4$$

$$A = \frac{4}{\left(\frac{k}{m}\right)} = \frac{4 \times 6}{48} = 0.50$$

PART-1 : PHYSICS

SECTION-II (ii)

6. Ans (248)

$$f_{\text{incident}} = \left(\frac{330 + 20}{330 - 50} \right) 1000$$

$$\left(\frac{350}{280} \right) 1000 = 1250$$

Speed of wave will be constant

$$\lambda_{\text{reflected}} = \frac{(330 - 20)}{1250} = 248 \text{ mm}$$

$$f_1 = \left(\frac{330}{330 + 50} \right) 1000$$

$$f_2 = \left(\frac{330}{330 - 20} \right) 1250$$

$$f_1 = 868 \text{ Hz}$$

$$f_2 = \frac{330}{310} \times 1250 = 1330 \text{ Hz}$$

$$f_2 - f_1 = 462$$

7. Ans (3)

Using conservation of mechanical energy and equation of continuity

$$\{\rho 2Ax\}g \left(H - \frac{x}{2} \right) = \frac{1}{2} \rho 4AH \cdot v^2$$

after differentiation we can get for $x = 0$, $a = g/2$

8. Ans (5)

$$E = \frac{-GMm}{2R}$$

$$\frac{dE}{dt} = Fv = \frac{-GMm}{2R^2} \times \frac{dR}{dt}$$

$$\Rightarrow \Delta R = \frac{2FvR^2}{GMm} \Delta t$$

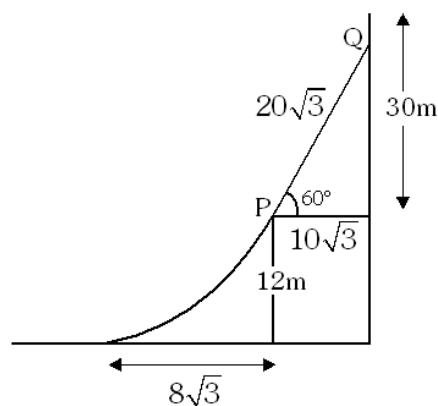
$$\frac{GM}{R^2} = g$$

$$\Rightarrow \Delta R = \frac{2Fv}{mg} \Delta t$$

$$= \frac{2 \times 0.01 \times \sqrt{10 \times 6400 \times 10^3} \times 24 \times 3600 \times 10}{3 \times 10^3 \times 10}$$

$$\Delta R = \frac{2 \times 24 \times 36 \times 8 \times 10^3}{3 \times 10^3 \times 10} = 4600.8 \text{ m} \approx 5 \text{ Km}$$

9. Ans (1800)



$$x^2 = 16y$$

$$2x = \frac{16dy}{dx}$$

$$\frac{dy}{dx} = \frac{x}{8} = \sqrt{3}$$

$$\theta = 60^\circ$$

$$W = \Delta K$$

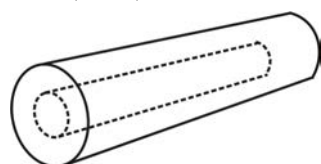
$$W_F + W_{ff} + W_g + W_N = 0$$

$$W_F = -W_{ff} - W_g$$

$$= \mu mg (8\sqrt{3} + 10\sqrt{3}) + mg(42)$$

$$= 1800 \text{ J}$$

10. Ans (126)



$$\frac{2000}{\pi r^2 \ell} \times \pi x^2 \ell = k \times 2\pi x \ell \frac{dT}{dx}$$

$$\frac{2000}{2\pi r^2 \ell k} \int x dx = - \int_T^{105} dT$$

$$\frac{2000}{2\pi \times 16 \times 10^{-6} \times 0.5 \times 15} \times \frac{1}{2} \times 16 \times 10^{-6} = T - 105$$

$$\frac{200}{3\pi} + 105 = T$$

$$T = 126.22 \approx 126^\circ \text{C}$$

$$= \frac{400}{3\pi} + 105 = T$$

$$\Rightarrow T = 126.22 \approx 126^\circ \text{C}$$

PART-1 : PHYSICS

SECTION-III

1. Ans (8)

aQ

$$E = -\frac{a}{2} \frac{dB}{dt}$$

$$\tau_{(\text{per unit length})} = a[2\pi a \sigma]E$$

$$\tau = I \frac{d\omega}{dt}$$

$$\Rightarrow d\omega = -\frac{\sigma \pi a^3}{I} \frac{dB}{dt}$$

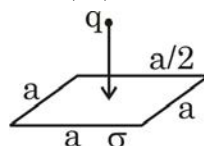
$$\omega = \frac{\sigma \pi a^3}{I} [B_i - B_f]$$

$$\text{Now } B_i = B$$

$$B_f = \mu_0 n i = \mu_0 a \sigma \omega$$

$$\Rightarrow \omega = \frac{\sigma \pi a^3 B}{I} \left[1 + \frac{\mu_0 \sigma^2 a^4 \pi}{I} \right]^{-1}$$

2. Ans (3)



$$\text{Flux due to } q = \int E \cdot dA = \frac{q}{6\epsilon_0}$$

$$\text{Force due to } q = \sigma \int \vec{E} \cdot d\vec{A} = \frac{\sigma q}{6\epsilon_0}$$

$$\Rightarrow \frac{\text{Force on } q}{q} = E = \frac{\sigma}{6\epsilon_0}$$

$$E_{\text{net}} = \frac{\sigma}{3\epsilon_0} \sqrt{1^2 + 1^2 + 1^2} = \frac{\sigma}{\sqrt{3}\epsilon_0}$$

3. Ans (3)

For floating

$$mg = \text{Buoyant force}$$

Since mass does not change with temperature

$$F_{B1} = F_{B2} \Rightarrow \rho_{w1} V_1 g = \rho_{w2} V_2 g$$

$$\rho_{w1} (0.99 V_0) = \left(\frac{\rho_{w1}}{1 + \gamma_w \Delta T} \right) (V_0 (1 + 3\alpha \Delta T))$$

$$\Rightarrow \left(1 - \frac{1}{100} \right) = \left(\frac{1 + 3\alpha \Delta T}{1 + \gamma_w \Delta T} \right)$$

By Binomial approx.

$$1 - \frac{1}{100} = (1 + (3\alpha - \gamma_w) \Delta T)$$

$$\frac{1}{100} = (\gamma_w - 3\alpha) \Delta T$$

$$\Rightarrow \Delta T = \frac{1}{100(\gamma_w - 3\alpha)} = 40^\circ \text{C}$$

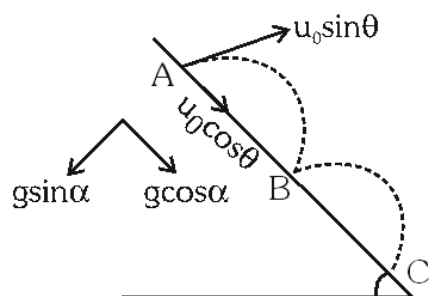
$$\Rightarrow T_f = 60^\circ \text{C}$$

4. Ans (3)

$$1 + \epsilon' r_A = 2 + 0.5 r_A \Rightarrow r_A = 2\Omega$$

$$\Rightarrow V = 1 + 2 = 3 \text{ volts}$$

5. Ans (5)



$$T_{A \rightarrow B} = \frac{2u_0 \sin \theta}{g \sin \alpha} = T_1$$

$$T_{B \rightarrow C} = \frac{2(u_0/\beta) \sin \theta}{g \sin \alpha} = \frac{T_1}{\beta} = T_2$$

$$\vdots \quad \vdots \quad \vdots \quad \vdots$$

$$s_{A \rightarrow B} = u_0 \cos \theta T_1 + \frac{1}{2} g \cos \alpha T_1^2$$

$$s_{B \rightarrow C} = \frac{u_0}{\alpha} \cos \theta T_2 + \frac{1}{2} g \cos \alpha T_2^2$$

$$= \frac{u_0}{\beta} \cos \theta \frac{T_1}{\beta} + \frac{1}{2} g \cos \alpha \frac{T_1^2}{\beta^2}$$

$$\vdots \quad \vdots \quad \vdots \quad \vdots$$

$$s_{\text{total}} = u_0 \cos \theta T_1 \left[1 + \frac{1}{\beta^2} + \frac{1}{\beta^4} + \dots \infty \right] +$$

$$\frac{1}{2} g \cos \alpha T_1^2 \left[1 + \frac{1}{\beta^2} + \frac{1}{\beta^4} + \dots \infty \right]$$

$$s_{\text{total}} = u \cos \theta T_1 \left[\frac{1}{1 - \frac{1}{\beta^2}} \right] + \frac{1}{2} g \cos \alpha T_1^2 \left[\frac{1}{1 - \frac{1}{\beta^2}} \right]$$

$$(V_{\text{avg}}) = \frac{s_{\text{total}}}{T_{\text{total}}} = \frac{\frac{u_0 \cos \theta T_1 \beta^2}{\beta^2 - 1} + \frac{g \sin \alpha T_1^2 \beta^2}{2(\beta^2 - 1)}}{T_1 \left[\frac{\beta}{\beta - 1} \right]}$$

6. Ans (4)

$$\frac{\Delta y}{y} \times 100 = \frac{2\Delta d}{d} \times 100 + \frac{\Delta \ell}{\ell} \times 100$$

$$= \frac{2 \times 5 \times 10^{-3} \times 100}{5 \times 10^{-1}} + \frac{5 \times 10^{-3} \times 100}{250 \times 10^{-3}}$$

$$= 2 + 2 = 4$$

7. Ans (2)

Let each side of the pendulum is displaced by x , the total extension of the spring become $2x$, where $x = \ell \sin \theta$. For small θ , $x = \ell \theta$.

$$\therefore \text{Force in the spring } F = k(2x) = 2k\ell\theta$$

$$\text{Restoring torque } \tau = -(mg \times \ell \sin \theta + 2k\ell\theta \times \ell \cos \theta)$$

$$\text{for small } \theta, \sin \theta = \theta \text{ and } \cos \theta = 1$$

$$\therefore \tau = -(mg\ell\theta + 2k\ell^2\theta)$$

$$\text{and } \alpha = \frac{(mg\ell + 2k\ell^2)}{I}(-\theta)$$

Compare with standard equation of SHM,

$$\alpha = -\omega^2 \theta \text{ we get}$$

$$\omega = \sqrt{\frac{mg\ell + 2k\ell}{I}}$$

$$\text{and } T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{I}{(mg\ell + 2k\ell^2)}}$$

$$= 2\pi \sqrt{\frac{m\ell^2}{mg\ell + 2k\ell^2}}$$

$$= 2\pi \sqrt{\frac{\ell}{\left(g + \frac{2k\ell}{m}\right)}}$$

8. Ans (6)

Let potential of point A is x and potential of point B is zero. Consider charge flown through 3V battery is q_0 .

$$2(3 - x) + q_0 + (0 - x)2 = 0 \quad \dots(1)$$

$$-q_0 - (x - 3) \times 1 + (2 - x - 3)2 = 0 \quad \dots(2)$$

On solving $x = 1$

$$q_0 = 2\mu C$$

$$w_D = q_0 \epsilon_1 = 6\mu J$$

PART-2 : CHEMISTRY

SECTION-II (i)

1. Ans (7.00)

III, VI, VII, $\rightarrow$ Phenolic ether will not give cleavage with HI.

3. Ans (24.63)

$$Z = \frac{PV}{nRT}$$

$$\frac{V}{n} = \frac{ZRT}{P} = \frac{0.8 \times 0.0821 \times 300}{8}$$

$$= 2.463 \text{ litre/mole}$$

∴ Volume of 10 mol of the gas = 24.63 litre.

4. Ans (45.15)



a 0

a - x 2x

$$\frac{2x}{a-x} = 2, \quad x = \frac{a}{2}$$

$$t = \frac{2.303}{92.12 \times 10^{-2}} \log \frac{a}{a-x}$$

$$t = \frac{2.303}{92.12 \times 10^{-2}} \log 2$$

$$t = \frac{2.303 \times 0.3010}{92.12 \times 10^{-2}}$$

$$t = 0.7525 \text{ min} = 45.15 \text{ sec.}$$

5. Ans (15.05)

Complex	Magnetic moment
$[\text{Cu}(\text{NH}_3)_4]^{2+}$	$\sqrt{3} = 1.73 \text{ B. M.}$
$[\text{Fe}(\text{CN})_6]^{3-}$	$\sqrt{3} = 1.73 \text{ B. M.}$
$[\text{NiCl}_4]^{2-}$	$2\sqrt{2} = 2.82 \text{ B. M.}$
$[\text{CoF}_6]^{3-}$	$\sqrt{24} = 4.90 \text{ B. M.}$
$[\text{Fe}(\text{H}_2\text{O})_5(\text{NO})]\text{SO}_4$	$\sqrt{15} = 3.87 \text{ B. M.}$

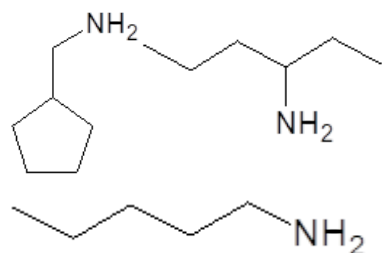
PART-2 : CHEMISTRY

SECTION-II (ii)

6. Ans (6)

Refer theory

7. Ans (3)



8. Ans (105)

$$\text{Rank of anion} - A = \frac{6}{8} = \frac{3}{4}$$

$$\text{Rank of anion} - B = \frac{2}{8} = \frac{1}{4}$$

$$\text{Rank of cation} - Z = 1 + \frac{6}{2} = 4$$

Hence ionic compound empirical formula is



$$d = \frac{\left(\frac{6}{8} \times 20\right) + \left(\frac{2}{8} \times 40\right) + (4 \times 15)}{(1)^3} \frac{\text{amu}}{\text{\AA}^3}$$

$$d = 85$$

$$\text{So, } a + b + c + d = 3 + 1 + 16 + 85 = 105$$

10. Ans (9)

White colour/colourless sp : 4, 6, 7, 8, 15, 16.

PART-2 : CHEMISTRY

SECTION-III

3. Ans (6)

$$i = 1.25$$

Original mole fraction

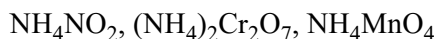
$$= \frac{1}{n} = \frac{1}{1 + (n-1)} = \frac{1.25}{1.25 + (n-1)} = \frac{1}{5} \Rightarrow n = 6$$

4. Ans (2)

$$\lambda = K \times \frac{1000}{M} = \frac{1}{100} \times \frac{20}{A} \times \frac{1000}{\frac{2 \times 1000}{20 \times A}}$$

$$= \frac{20}{100A} \times \frac{20A}{2} = 2 \text{ Scm}^2\text{mol}^{-1}$$

6. Ans (3)



7. Ans (9)

$$x = 2, y = 3, z = 4$$

8. Ans (8)



PART-3 : MATHEMATICS

SECTION-II (i)

1. **Ans (0.50)**

$$\text{Let } \frac{x}{2^{r+1}} = \theta$$

$$T_r = \tan \theta \frac{(1 + \tan^2 \theta)}{(1 - \tan^2 \theta)} = \frac{\sin \theta}{\cos 2\theta \cdot \cos \theta}$$

$$T_r = \frac{\sin(2\theta - \theta)}{\cos 2\theta \cos \theta} = \tan 2\theta - \tan \theta$$

$$T_r = \tan\left(\frac{x}{2^r}\right) - \tan\left(\frac{x}{2^{r+1}}\right)$$

$$T_0 = \tan(x) - \tan\left(\frac{x}{2}\right)$$

$$T_1 = \tan\left(\frac{x}{2^4}\right) - \tan\left(\frac{x}{2^2}\right)$$

⋮

$$S_n = \tan x \Rightarrow f(x) = \tan x$$

2. **Ans (0.40)**

Equation of plane through A, B, C is

$$x - 5y - 2z + 6 = 0$$

Equation of perpendicular from P to the plane is

$$\frac{x-4}{1} = \frac{y+\frac{26}{3}}{-5} = \frac{z+\frac{10}{3}}{-2} = \lambda$$

$$x = 4 + \lambda, y = -5\lambda - \frac{26}{3}, z = -2\lambda - \frac{10}{3}$$

$$\text{Point of intersection is } \left(2, \frac{4}{3}, \frac{2}{3}\right)$$

3. **Ans (0.25)**

$$T_r = \tan^{-1} \left(\frac{7^r - 7^{(r-1)}}{1 + 7^r \cdot 7^{(r-1)}} \right)$$

$$T_r = \tan^{-1}(7^r) - \tan^{-1}(7^{(r-1)})$$

$$S_\infty = \frac{\pi}{4}$$

4. **Ans (2.00)**

$$(\sec A \sec B + \tan A \tan B)^2 - (\sec A \tan B + \tan A \sec B)^2$$

$$= \frac{(1 + \sin A \sin B)^2 - (\sin B + \sin A)^2}{\cos^2 A \cos^2 B}$$

$$= \frac{1 + \sin^2 A \sin^2 B - \sin^2 B - \sin^2 A}{\cos^2 A \cos^2 B} = 1$$

$$(\sec A \sec B + \tan B \tan A)^2 = (91)^2 + 1 = 8282$$

5. **Ans (2.00)**

Minimum distance is along their common normal.

PART-3 : MATHEMATICS

SECTION-II (ii)

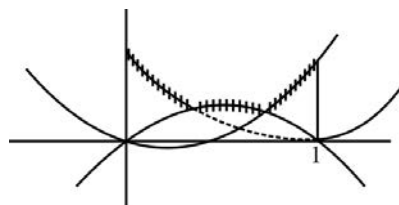
6. **Ans (18)**

$$7(7^2)^{51} = 7(50 - 1)^{51}$$

$$= 7({}^{51}C_0(50)^{51} - {}^{51}C_1(50)^{50} + \dots + (-1))$$

last is -7 but remainder is 18

7. **Ans (17)**



$$\int_0^{1/3} (1-x)^2 dx + \int_{1/3}^{2/3} 2x(1-x) dx + \int_{2/3}^1 x^2 dx = \frac{17}{27}$$

8. **Ans (40)**

Solving the integral, we get

$$\frac{a^3}{3} - a^2b + 3ac = 0 \quad \text{div. by } a \quad \text{we get}$$

$$\frac{a^2}{3} - ab + 3c = 0$$

since $-ab + 3c$ is an integer then a must be divisible by 3 i.e. $a = 3k$ so that the expression is an integer

so, options are,

$$a = 3, b = 2, c = 1$$

$$a = 6, b = 3, c = 2$$

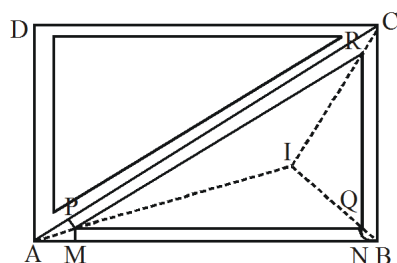
$$a = 9, b = 4, c = 3$$

$$\text{Total cases} = {}^{10}C_3 = 120.$$

$$\text{required prob} = \frac{3}{120} = \frac{1}{40} = \frac{1}{p} \therefore P = 40.$$

9. Ans (59)

Since the circle lies within the rectangle, therefore centre of circle must lie within a rectangle of size $(9 - 2) \times (12 - 2)$ or 7×10 . Also, circle does not touch and intersect the diagonal AC, so the distance from the centre of the circle to the diagonal AC and each side of $\triangle ABC$ and $\triangle ACD$ is greater than 1.



so, required probability

$$= \frac{2 \cdot \text{ar}(\triangle PQR)}{\text{area of rectangle of size } 7 \times 10}$$

$$= \frac{2 \cdot \text{ar}(\triangle PQR)}{\text{ar}(\triangle ABC)} \cdot \frac{\text{ar}(\triangle ABC)}{7 \times 10} = \frac{PQ^2}{AB^2} \cdot \left(\frac{9 \times 12}{7 \times 10} \right)$$

($\because \triangle ABC \sim \triangle PQR$)

$$= \frac{54}{35} \cdot \frac{MN^2}{AB^2}$$

Let $A(0,0)$, $B(12,0)$ & $C(12,9)$, then the slope

of $AI = \frac{1}{3}$ (where I is incentre)

Also, $PM = AN = BN = 1$, so $AM = 3$

$$\Rightarrow MN = 8.$$

$\therefore$ req. probability

$$= \frac{54}{35} \cdot \frac{8^2}{12^2} = \frac{54}{35} \cdot \frac{4}{9} = \frac{24}{35} = \frac{m}{n}$$

$$\therefore m + n = 59.$$

10. Ans (36)

$$S_n = \int_0^1 x^2(1-x)^n dx = \int_0^1 (1-x)^2 \cdot x^n dx$$

(By king property)

$$= \int_0^1 (x^n - 2x^{n+1} + x^{n+2}) dx$$

$$\Rightarrow S_n = \left(\frac{x^{n+1}}{n+1} - \frac{2x^{n+2}}{n+2} + \frac{x^{n+3}}{n+3} \right)_0^1$$

$$\Rightarrow S_n = \frac{1}{n+1} - \frac{2}{n+2} + \frac{1}{n+3}$$

$$= \left(\frac{1}{n+1} - \frac{1}{n+2} \right) - \left(\frac{1}{n+2} - \frac{1}{n+3} \right)$$

$$\therefore \sum_{k=1}^n S_k = \left(\frac{1}{2} - \frac{1}{n+2} \right) - \left(\frac{1}{3} - \frac{1}{n+3} \right)$$

$$\Rightarrow \lim_{n \rightarrow \infty} \sum_{k=1}^n S_k = \frac{1}{2} - \frac{1}{3} = \frac{1}{6} = \frac{1}{\sqrt{36}} \therefore A = 36$$

PART-3 : MATHEMATICS

SECTION-III

1. Ans (1)

$$T_n = \frac{(n+2)}{n(n+1)} \left(\frac{1}{2} \right)^n$$

$$= \left(\frac{2(n+1)-n}{n(n+1)} \right) \left(\frac{1}{2} \right)^n$$

$$T_n = \frac{1}{n} \left(\frac{1}{2} \right)^{(n-1)} - \frac{1}{(n+1)} \left(\frac{1}{2} \right)^n$$

$$\text{Sum} = 1 - \frac{1}{(n+1)2^n}$$

$$S_\infty = 1$$

2. Ans (1)

$$(1+x)^{10} = a_0 + a_1x + a_2x^2 + \dots + a_{10}x^{10}$$

put $x = i$ and solve

3. Ans (1)

$$f(x+y) = f(x) + f(y) - xy - 1$$

$$x = y = 1, f(2) = 0$$

$$x = 2, y = 1, f(3) = -2$$

$$f(n+1) = f(n) - n$$

$$f(n) - f(n+1) > 0$$

$$f(n) > f(n+1)$$

$$\text{Possible only } f(1) = 1$$

4. **Ans (2)**

Plane containing the line is

$$(x + y + 2z - 3) + \lambda(2x + 3y + 4z - 4) = 0$$

$$x(2\lambda + 1) + y(3\lambda + 1) + z(4\lambda + 2) - (4\lambda + 3) = 0$$

It's DC's are $((2\lambda + 1), (3\lambda + 1), (4\lambda + 2))$ & z-axis as $(0, 0, 1)$

$$\lambda = -\frac{1}{2}$$

$$y + z = 0$$

Distance of this plane from $(0, 0, 0)$ which lies on z-axis is 2

5. **Ans (1)**

$$\frac{d}{dx}(y^2 \sin x) = \sin 2x$$

$$y^2 \sin x = \frac{-\cos(2x)}{2} + C$$

$$\text{at } x = \frac{\pi}{2}, C = \frac{1}{2}$$

$$y^2 \sin x = \frac{1}{2}(1 - \cos 2x)$$

$$\frac{y^2}{\sin x} = 1$$

6. **Ans (1)**

$$(1)^\infty$$

$$\lim_{x \rightarrow 0} e^{\frac{2}{x}} \left(\frac{a^x + b^x + c^x}{3} - 1 \right)$$

$$\lim_{x \rightarrow 0} e^{\frac{2}{x}(\ell na + \ell nb + \ell nc)} = (abc)^{2/3}$$

7. **Ans (2)**

$$A^2 = 0, A^3 = 0 \dots$$

$$(A + I)^{50} = {}^{50}C_0 I^{50} + {}^{50}C_1 A I^{49} + 0 + \dots$$

$$(A + I)^{50} - 50A = I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$a + b + c + d = 2$$

8. **Ans (1)**

$$|z_1^2 - z_2^2| = |\bar{z}_1^2 + \bar{z}_2^2 - 2\bar{z}_1 \bar{z}_2|$$

$$|z_1^2 - z_2^2| = |z_1^2 + z_2^2 - 2z_1 z_2|$$

$$|z_1 + z_2| |z_1 - z_2| = |z_1 - z_2| |z_1 - z_2|$$

$$\Rightarrow |z_1 + z_2| = |z_1 - z_2|$$

$$\Rightarrow z_1 \perp z_2$$

$$\Rightarrow \arg\left(\frac{z_1}{z_2}\right) = 2n\pi + \frac{\pi}{2}$$

$$\Rightarrow -\frac{\pi}{2}, \frac{\pi}{2}, \frac{3\pi}{2}$$

minimum value of $|a + b| = 1$, ($a = -1$ & $b = 2$)



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE (Advanced)

FULL SYLLABUS

27-04-2024

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE ADVANCED)

ANSWER KEY

PAPER-2 (OPTIONAL)

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A,D	A,B,C	A,D	A,D	A,C	B,C,D
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	C	A	D	A	B	C
SECTION-II	Q.	1	2	3	4	5	6
	A.	2434	1324	1432	2234	1432	2133

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A,B,C,D	A,B,D	A,B,D	C,D	A,D	B,C,D
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A	D	D	A	C	B
SECTION-II	Q.	1	2	3	4	5	6
	A.	2143	4132	3241	3241	4321	2341

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	C,D	A,B,D	A,B,C	A,C	A,B,C,D	C,D
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	C	A	D	A	B	A
SECTION-II	Q.	1	2	3	4	5	6
	A.	2341	2132	4213	1143	3124	2431

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (A,D)

$$KE_{\max} = (5 - \phi) \text{ eV}$$

when these electrons are accelerated through 5V,

they will reach the anode with maximum energy = $(5 - \phi + 5) \text{ eV}$

$$\therefore 10 - \phi = 8$$

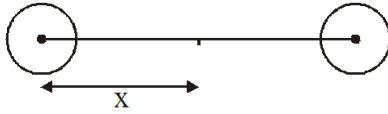
$$\phi = 2 \text{ eV Ans.}$$

Current is less than saturation current because if slowest electron also reached the plate it would have 5eV energy at the anode, but there it is given that the minimum energy is 6eV.

3. Ans (A,D)

The two wires are at P.D. V

So Let charges on them is Q and $-Q$ there cross section



$$E = \frac{Q}{2\pi\epsilon_0 x l} + \frac{Q}{2\pi\epsilon_0 (\eta r - x) l}$$

$$V = \int E \cdot dr = \frac{Q}{2\pi\epsilon_0 l} \left[\ln \eta - \ln \frac{1}{\eta} \right]$$

$$V = \frac{Q \ln \eta}{\pi\epsilon_0 l}$$

$$\Rightarrow \lambda = \frac{Q}{l} = \frac{\pi\epsilon_0 V}{\ln \eta}$$

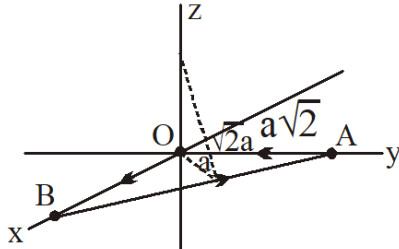
for zero force magnetic repulsion must balance attraction.

$$\frac{Q}{2\pi\epsilon_0 \eta r l} \left(\frac{Q}{l} \right) = \frac{\mu_0 V^2}{2\pi R \eta r}$$

$$\frac{1}{2\pi\epsilon_0 \eta r} \frac{\pi^2 \epsilon_0^2 V^2}{(\ln \eta)^2} = \frac{\mu_0 V^2}{2\pi R^2 \eta r}$$

$$R = \sqrt{\frac{\mu_0}{\epsilon_0} \frac{\ln \eta}{\pi}}$$

4. Ans (A,D)



$$\vec{B}_1 = \frac{\mu_0 I}{4\pi a} \left[\vec{O} + \frac{\sqrt{2}}{\sqrt{3}} \right] (-\hat{i}) = \frac{\mu_0 I}{4\pi a} \frac{\sqrt{2}}{\sqrt{3}} (-\hat{i})$$

$$= \frac{\mu_0 I}{2\sqrt{6}\pi a} (-\hat{i})$$

$$\vec{B}_2 = \frac{\mu_0 I}{4\pi a} \frac{\sqrt{2}}{\sqrt{3}} (-\hat{j}) = \frac{\mu_0 I}{2\sqrt{6}\pi a} (-\hat{j})$$

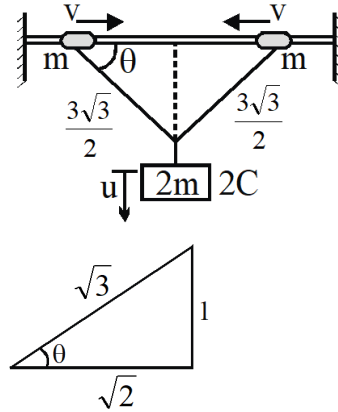
5. Ans (A,C)

$$v = \frac{F_0}{\sqrt{\left(\omega m - \frac{2k}{\omega}\right)^2 + b^2}} \cos(2\pi t - \phi)$$

$$\tan \phi = \frac{\omega m - \frac{2k}{\omega}}{b} = \frac{10\pi - \frac{200}{2\pi}}{b} = 0$$

$$\Rightarrow v_0 = \frac{F_0}{b}$$

6. Ans (B,C,D)



$v = u \tan \theta$ (constraint relation)

$$2mg \cdot \frac{3\sqrt{3}}{2} \sin \theta = 2 \times \frac{1}{2} m (u \tan \theta)^2 + \frac{1}{2} 2mu^2$$

$$u^2 = 3\sqrt{3} g \sin \theta \cos^2 \theta$$

To maximize u

$$\frac{du^2}{d\theta} = 0$$

$$\tan \theta = \frac{1}{\sqrt{2}}$$

$$u = \sqrt{20}$$

When load has maximum speed, speed of sleeve is not maximum as function is different.

$$\text{Descent of load} = \frac{3\sqrt{3}}{2} \sin \theta = \frac{3}{2} = 1.5 \text{ m}$$

Option D is correct as horizontal forces are zero and vertical forces are zero at the instant of maximum speed of load.

PART-1 : PHYSICS

SECTION-I (ii)

7. Ans (C)

Use basic concepts

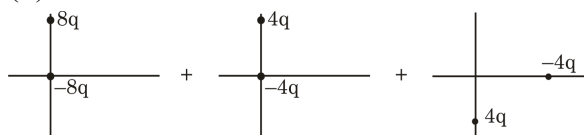
$$\Rightarrow Q = CV \Rightarrow C_{\text{eff in series}} = \frac{C_1 C_2}{C_1 + C_2}$$

$$\Rightarrow E = \frac{1}{2} CV^2$$

$$\text{PD in series is distributed as } \frac{1}{C_1} : \frac{1}{C_2} \Rightarrow C = \frac{\epsilon_0 A}{d}$$

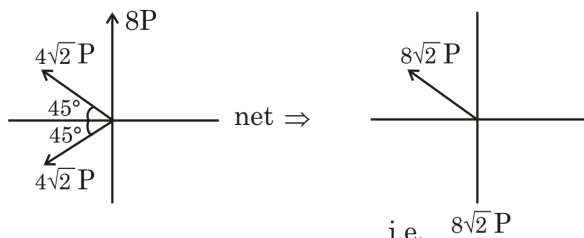
8. Ans (A)

(P) Consider it as

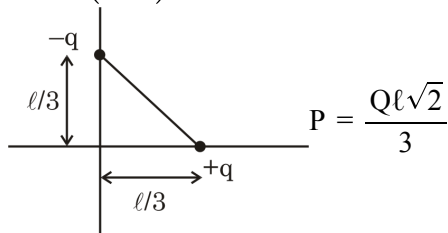


Let $P = Ql$

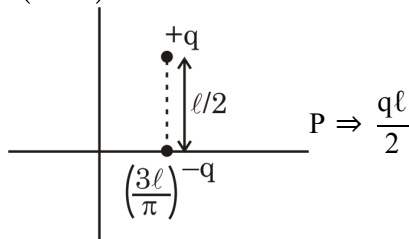
than we have



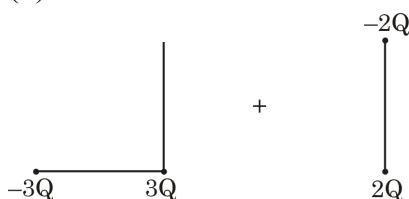
(Q) We can assume charge of cone to be at point $(0, \frac{1}{3})$



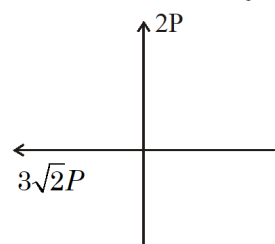
(R) We can assume charge $-q$ at point $(\frac{3l}{\pi}, 0)$



(S) Assume as



We have $P = Ql$



Let $= \sqrt{22} P$

$\sqrt{22} Ql$

9. Ans (D)

(A) Let $R_2 = x$ and $R_1 = 4x$

Now, $\Delta V_{(R_1)} = (0.1) \times 4x$

$\Rightarrow \Delta V_{(R_2 \& \text{Lamp})} = -0.1 \times 4x = -0.4x$

$\Rightarrow \Delta i_{(R_2)} = -0.4$

$\Rightarrow \Delta i_{\text{Lamp}} = 0.5$ as $\Rightarrow \Delta i_{(R_2)} + \Delta i_{\text{Lamp}} = 0.1 \text{ A}$

(B) let I be the impulse

For bar $I = Mv$; $\frac{IL}{2} = \frac{ML^2}{12} \omega \Rightarrow \frac{6v}{L} = \omega$

For puck $I = mu$

Also $u = v + \frac{L}{2} \omega = 4v$

So, $\frac{M}{m} = \frac{u}{v} = 4$

(C) Two possibilities $l = 20 \text{ cm}$ or $l = 80 \text{ cm}$

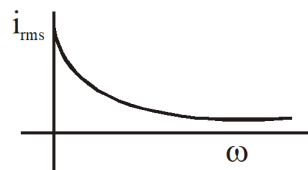
$\frac{R}{X} = \frac{\ell}{100 - \ell}$

(D) By symmetry, $\int_{-\infty}^{\infty} \vec{B} \cdot d\vec{y} = \frac{1}{2}(\mu_0 i) \Rightarrow n = \frac{1}{2}$

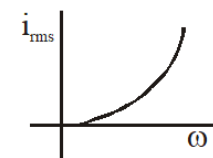
10. Ans (A)

For $R - L$ circuit, Impedence

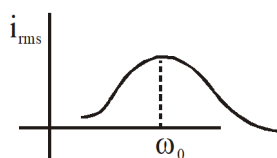
$Z = \sqrt{R^2 + (\omega L)^2}$ $i_{\text{rms}} = \frac{V_{\text{rms}}}{\sqrt{R^2 + \omega^2 L^2}}$



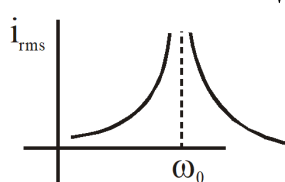
for $R - C$ circuit $i_{\text{rms}} = \frac{V_{\text{rms}}}{\sqrt{R^2 + \frac{1}{\omega^2 C^2}}}$



For $L - C - R$ circuit $i_{\text{rms}} = \frac{V_{\text{rms}}}{\sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2}}$



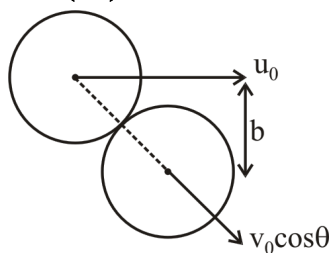
For $L - C$ circuit $i_{\text{rms}} = \frac{V_{\text{rms}}}{\sqrt{(\omega L - \frac{1}{\omega C})^2}}$



11. Ans (B)

Due to centrifugal force clockwise torque will increase to counter this N_2 will increase.

12. Ans (C)



Velocity along normal gets exchanged

$$\sin \theta = \frac{b}{2r} = \frac{12}{20} = \frac{3}{5}$$

$$\theta = 37^\circ$$

$$\Rightarrow v_b = v_0 \cos 37 = 10 \times \frac{4}{5} = 8 \text{ m/s}$$

$$\sin \theta = \frac{16}{20} = \frac{4}{5}$$

$$\theta = 53^\circ$$

$$u_b = v_0 \cos 53 = 10 \times \frac{3}{5} = 6 \text{ m/s}$$

PART-1 : PHYSICS

SECTION-II

1. Ans (2434)

Let m_x, m_y be nuclear masses of x and y

$$A. m_x - m_y - m_\alpha = (M_x - zm_e) -$$

$$[M_y - (z - 2)m_e] - (M_{He} - 2m_e)$$

$$B. m_x - m_y - m_e = (M_x - zm_e) -$$

$$[M_y - (z + 1)m_e] - m_e = M_x - M_y$$

$$C. m_x - m_y - m_e = (M_x - zm_e) -$$

$$[M_y - (z - 1)m_e] - m_e = M_x - M_y - 2m_e$$

$$D. (m_e + m_x) - m_y = m_e + (M_x - zm_e) -$$

$$[M_y - (z - 1)m_e] = M_x - M_y \text{ (K-capture)}$$

2. Ans (1324)

Critical angle for total internal reflection, $\theta_c \propto \lambda$
[Cauchy's formula and T.I.R. concept]

$$(\theta_c)_{\text{Red}} > (\theta_c)_{\text{Yellow}} > (\theta_c)_{\text{Green}} > (\theta_c)_{\text{blue}}$$

So, red & yellow colour will undergo T.I.R. later as compared to blue & green.

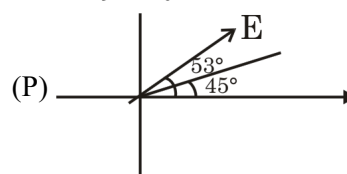
And, $(r)_A < (r)_B$. So, A is Red & B is yellow.

Again with the same concept, D is blue colour and c is green.

3. Ans (1432)

$$I_0 = \frac{1}{2} C \epsilon_0 E_0^2 = \frac{1}{2} \times 3 \times 10^8 \times 4\pi \times 5^2 \epsilon_0$$

$$= \frac{3 \times 25 \times 10^8}{2 \times 9 \times 10^9} = \frac{5}{12} \text{ w/m}^2$$



$$I = I_0 \cos^2 (53 - 45) = \frac{5}{12} \left[\frac{3}{5} \times \frac{1}{\sqrt{2}} + \frac{4}{5} \times \frac{1}{\sqrt{2}} \right]^2$$

$$= \frac{5}{12} \times \frac{49}{25} \times \frac{1}{2} = \frac{49}{120} \text{ w/m}^2$$

$$(Q) I = I_0 \cos^2 37^\circ = \frac{4}{15} \text{ w/m}^2$$

$$(R) I = I_0 \cos^2 53^\circ = \frac{3}{20} \text{ w/m}^2$$

$$(S) I = I_0 \cos^2 (53 - 37) = \frac{48}{125} \text{ w/m}^2$$

4. Ans (2234)

$$P_b - P_a = 0,$$

$V_a \ll V_b \Rightarrow V_a - V_b < 0$, pressure in the tube must be non-zero everywhere

$$\therefore P_0 > (h_1 + h_2)\rho g, V_b \approx \sqrt{2gh} \Rightarrow h_1 > 0$$

$$P_0 > (h_1 + h_2)\rho g, V_b \approx \sqrt{2gh} \Rightarrow h_1 > 0$$

5. Ans (1432)

$$P_0 V_0 = P(V_0 + Avt)$$

$$P = \frac{P_0 V_0}{V_0 + Avt}$$

$$F_{\text{agent}} = (P_0 - P)A = P_0 \left[1 - \frac{V_0}{V_0 + Avt} \right] A$$

$$= \frac{P_0 A^2 vt}{V_0 + Avt}$$

$$W = P_0 V_0 Av \ln \left(\frac{V_0 + Avt}{V_0} \right)$$

$$W = P_0 V_0 Av \ln \frac{V}{V_0}$$

6. Ans (2133)

$$\vec{V}_1 = 2\vec{V}_m - \vec{V}_0$$

$$(P) \vec{V}_1 = 2V - 2V = 0$$

$$(Q) V_{1/M} = V_1 - V_M = 0 - V$$

$$(R) V_{1/0} = V_1 - V_0 = -2V$$

(S) When spring max compressed

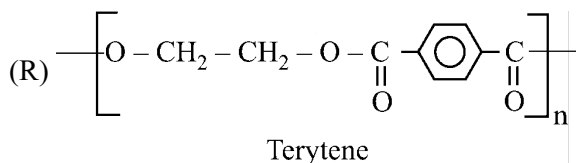
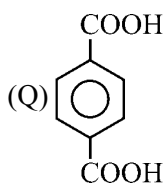
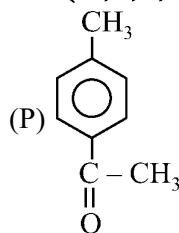
$$V_m = 0$$

$$V_1 = -V_0 = 2V$$

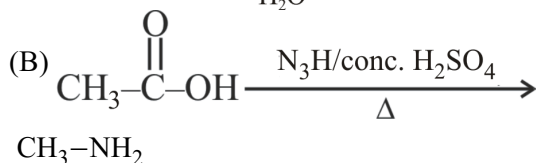
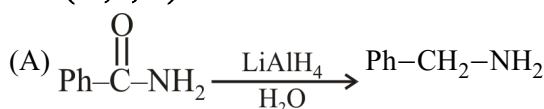
PART-2 : CHEMISTRY

SECTION-I (i)

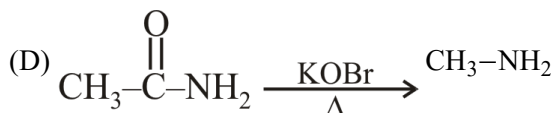
1. **Ans (A,B,C,D)**



2. **Ans (A,B,D)**



(C) Chlorobenzene doesn't give SN^2 reaction due to partial double bond character in between carbon & chlorine.



6. **Ans (B,C,D)**

(B) Rate constant increases with increases in temperature. This is due to a greater number of collisions whose energy exceeds the activation energy.

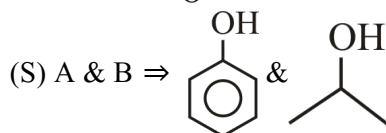
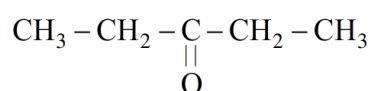
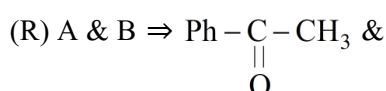
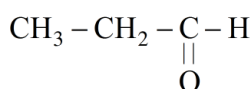
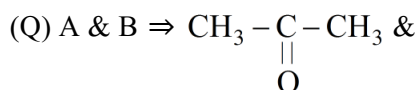
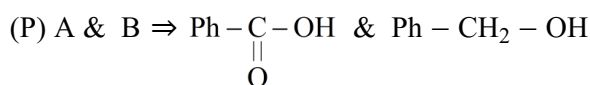
(C) Higher the magnitude of activation energy, stronger is the temperature dependence of the rate constant.

(D) The pre-exponential factor is a measure of the rate at which collisions occur, irrespective of their energy.

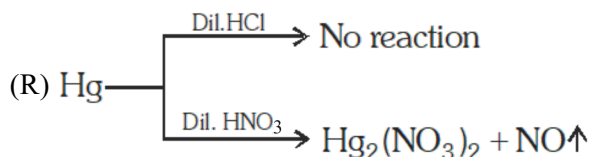
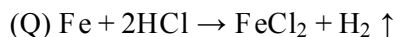
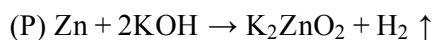
PART-2 : CHEMISTRY

SECTION-I (ii)

8. **Ans (D)**



9. **Ans (D)**



(S) Au and Pt do not react with HNO_3 or H_2SO_4 or HCl etc.

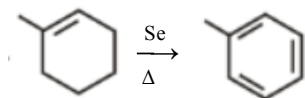
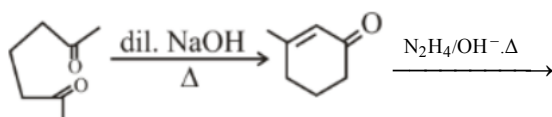
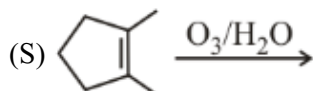
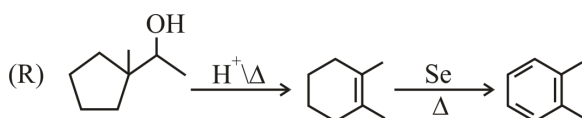
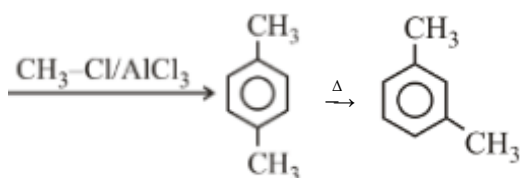
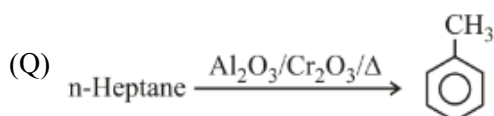
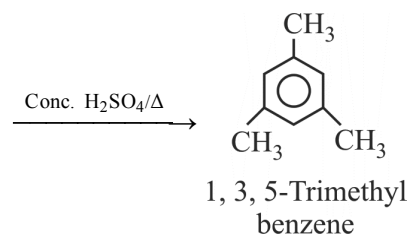
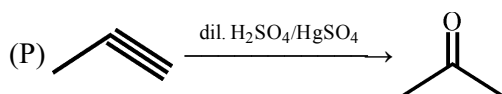
12. **Ans (B)**

When the solutions are mixed number of particles per unit volume decreases.

PART-2 : CHEMISTRY

SECTION-II

2. Ans (4132)



PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (C,D)

$$f(x) = (-n^3) \left(\left| 2\tan^{-1}(e^{|x|} - e) - \frac{1}{n^3} \right| - 2 \left| \tan^{-1}(e^{|x|} - e) \right| \right)$$

$$\text{Let } g(x) = e^{|x|} - e$$

$$= \lim_{n \rightarrow \infty} \frac{(-n^3) \left(\left(2\tan^{-1}g(x) - \frac{1}{n^3} \right)^2 - 4(\tan^{-1}g(x))^2 \right)}{\left| 2\tan^{-1}g(x) - \frac{1}{n^3} \right| + 2 \left| \tan^{-1}g(x) \right|}$$

$$= \lim_{n \rightarrow \infty} \frac{(-n^3) \left(\frac{-4\tan^{-1}g(x)}{n^3} + \frac{1}{n^6} \right)}{\left| 2\tan^{-1}g(x) - \frac{1}{n^3} \right| + 2 \left| \tan^{-1}g(x) \right|}$$

$$= \frac{\tan^{-1}g(x)}{\left| \tan^{-1}g(x) \right|} \quad x \neq \pm 1$$

$$f(x) = \begin{cases} \frac{\tan^{-1}g(x)}{\left| \tan^{-1}g(x) \right|} & x \neq \pm 1 \\ -1 & x = \pm 1 \end{cases}$$

(A) $f(x)$ is discontinuous $x = \pm 1$

(B) $|f(x)|$ is continuous function

2. Ans (A,B,D)

$$f(x) \left[(f(x))^6 + 1 \right] = x$$

$$f(0) \left[(f(0))^6 + 1 \right] = 0 \Rightarrow f(0) = 0$$

$$7(f(x))^6 f'(x) = 1 - f'(x)$$

$$f'(x) \left[7(f(x))^6 + 1 \right] = 1 \Rightarrow f'(x) > 0$$

$\therefore f(x)$ is increasing function

$$(f(x))^7 = x - f(x)$$

$$x \rightarrow f^{-1}(x)$$

$$x^7 = f^{-1}(x) - x \Rightarrow f^{-1}(x) = x^7 + x$$

3. Ans (A,B,C)

$$P \equiv 3x - 3y + z - 2 = 0$$

$$Q \equiv x - y + cz - 1 = 0$$

$$P \parallel Q \Rightarrow \frac{3}{1} = \frac{1}{c} \Rightarrow c = \frac{1}{3}$$

$$\cos 45^\circ = \frac{\vec{n}_1 \cdot \vec{n}_2}{|\vec{n}_1| |\vec{n}_2|} = \left| \frac{6+c}{\sqrt{19}\sqrt{2+c^2}} \right|$$

$$\Rightarrow 17c^2 - 24c - 34 = 0 \Rightarrow c_1 c_2 = -2$$

$$\rightarrow x_0 = 3, y_0 = 4, z_0 = 5$$

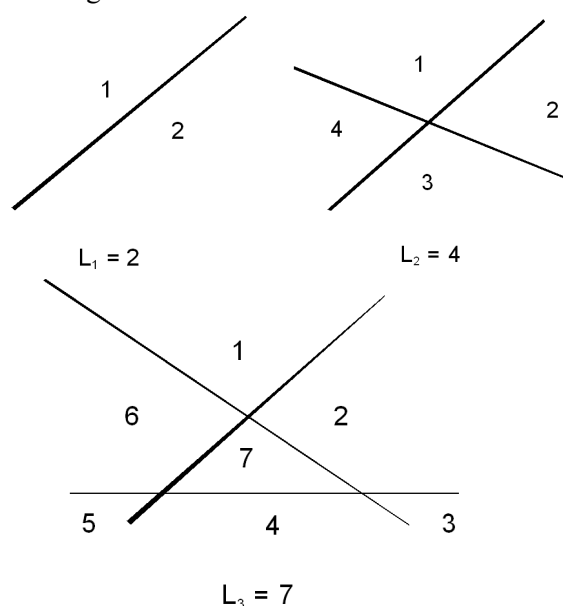
$$\Rightarrow x_0 + y_0 + z_0 = 12$$

4. Ans (A,C)

$$P = \frac{{}^5C_1 ({}^8C_2 - 4) \cdot ({}^6C_2 - 2)}{{}^{10}C_2 \cdot {}^8C_2 \cdot {}^6C_2} = \frac{26}{315} = \frac{m}{n}$$

5. Ans (A,B,C,D)

Clearly the number of regions is maximized when no two lines are parallel and no three are concurrent. Easy to see that $L_1 = 2$, $L_2 = 4$ and $L_3 = 7$. For the general case, we observe that when we add the n th line, this new line is cut by the previous $n - 1$ lines in $n - 1$ points and hence the new line is divided into n segments. Each of these segments divides a previous region into 2 parts and hence the number of new regions



Lines dividing a plane added when we introduce the n^{th} line is n . Thus we have the recurrence relation $L_n = L_{n-1} + n$. Hence $L_4 = 11$, $L_5 = 16$, ...

$$L_n - L_{n-1} = n$$

$$L_{n-1} - L_{n-2} = n-1$$

$$\dots \dots \dots$$

$$L_3 - L_2 = 4$$

$$L_2 - L_1 = 2$$

Adding, we get

$$L_n - L_1 = n + (n-1) + (n-2) + \dots + 2$$

$$= \frac{n(n+1)}{2} - 1$$

$$\text{Thus } L_n = \frac{n(n+1)}{2} - 1 + 2 \quad L_{10} = 56$$

$$= \frac{n(n+1)}{2} + 1 \quad L_{15} = 121$$

$$= \frac{1}{2}(n^2 + n + 2) \quad L_{20} = 211$$

$$L_{25} = 326$$

6. Ans (C,D)

Using characteristic equation $\lambda_1 = 2$, $\lambda_2 + \lambda_3 = 3$;

$$|A| = 2$$

We get $A^3 - 5A^2 + 7A - 2I = 0$

$$(A) \frac{1}{81} |\text{adj } 2A| (\text{adj } 3A^{-1}) |A^{-1}|$$

$$= \frac{1}{81} \times |2A|^2 |3A^{-1}| \frac{1}{|A|}$$

$$= \frac{1}{81} \times 2^6 \times 3^6 \times \frac{1}{2}$$

$$= 32 \times 9 = 288$$

$$(B) (A^{-1})^2 = \frac{1}{2} (A - 5I + 7A^{-1})$$

$$(C) |A^5 - 2A^4| = |A|^4 |A - 2I| = 0$$

As 2 is an eigen value

$$(D) 2A^{-1} = 7I + A^2 - 5A$$

$$A^{-1} = \frac{(A^2 - 5A + 7I)}{2}$$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. Ans (C)

(P) : Equation of the plane $A(x-1) + B(y-1) + C(z-1) = 0$

Since the line is perpendicular to the plane (1)

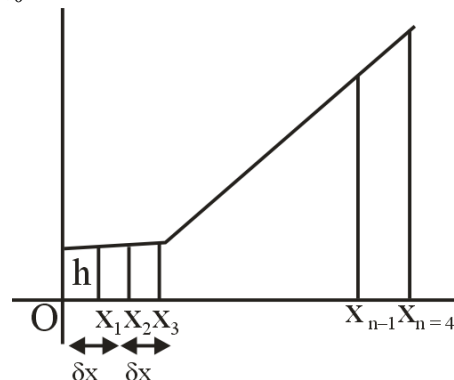
$$\therefore 3(x-1) + 0(y-1) + 4(z-1) = 0$$

Distance from $(0,0,0)$

$$d = \left| \frac{-7}{5} \right| = \frac{7}{5} = \frac{p}{q}$$

$$\Rightarrow p - q = 2$$

$$(Q) L = \lim_{\delta x \rightarrow 0} \delta x (x_1 + x_2 + x_3 + \dots + x_n)$$



$$= \lim_{n \rightarrow \infty} \left[\frac{4}{n} + \frac{8}{n} + \frac{12}{n} + \dots + 4 \cdot \frac{n}{n} \right] \left(\delta x = \frac{4}{n} \right)$$

$$= \lim_{n \rightarrow \infty} \frac{16}{n^2} (1 + 2 + 3 + \dots + n) = \lim_{n \rightarrow \infty} \frac{16}{n^2} \cdot \frac{n(n+1)}{2} = 8$$

(R) If x is replaced by $-x$ in the given equation, then

$$-x f(-x) + (1+x) f(x) = x^2 - x + 1$$

subtracting the two equation we get

$$f(-x) = f(x) + 2x,$$

subtracting the value of $f(-x)$ in the given equation

we get

$$xf(x) + (1 - x)(f(x) + 2x) = x^2 + x + 1 \text{ and thus}$$

$$f(x) = x^2 + x + 1 - 2x(1 - x) = 3x^2 - x + 1 = 3$$

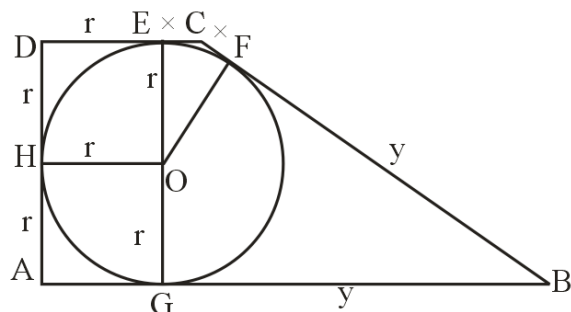
$$\left(x - \frac{1}{6}\right)^2 + \frac{11}{12}$$

and hence $f(x) \geq \frac{11}{12}$. If $x = \frac{1}{6}$ then we get

$$f\left(\frac{1}{6}\right) = \frac{11}{12} = \frac{p}{q}$$

$$\Rightarrow (q - p) = 1$$

(S) Using the notation from the above diagram and the conditions from the problem one obtains :



$$3(r+x)=r+y \Leftrightarrow y=3x+2r$$

$$\text{and } (x+y)^2 = 4r + \{(r+y) - (r+x)^2\} \Leftrightarrow xy = r^2$$

$$\text{Hence } x(3x + 2r) = r^2 \Leftrightarrow x = \frac{r}{3}$$

And consequently $y = 3r$. On the other hand the area of the trapezoid ABCD is 4, thus $(r + x + r + y)r = 4$.

Substituting for $x = \frac{r}{3}$ and $y = 3r$,

$$\text{we get } r = r = \frac{\sqrt{3}}{2} \Rightarrow 4r^2 = 3$$

8. Ans (A)

(P) $x + 2x + \dots + 12x = 78x$

$$\therefore \{x\} = x, \{2x\} = 2x, \dots, \{12x\} = 12x$$

$$\therefore 0 \leq 12x < 1 \Rightarrow 0 \leq x < \frac{1}{12}$$

Since $x \in \left[\frac{1}{15}, \frac{1}{10} \right]$

$$\therefore x \in \left[\frac{1}{15}, \frac{1}{12} \right) \Rightarrow \frac{1}{15}, \frac{1}{14}, \frac{1}{13}$$

$\therefore$ Number of values $x = 3$

$$(Q) (2x^2 + 3x + 4)^{10} = \sum_{r=0}^{20} a_r x^r \text{ replace } x \text{ by } \frac{2}{x}$$

we get

$$\left(\frac{8}{x^2} + \frac{6}{x} + 4\right)^{10} = \sum_{r=0}^{20} a_r \frac{2^r}{x^r}$$

$$\Rightarrow 2^{10} (4 + 3x + 2x^2)^{10} = \sum_{r=0}^{20} a_r 2^r x^{20-r}$$

$$\Rightarrow 2^{10} \sum_{r=0}^{20} a_r x^r = \sum_{r=0}^{20} a_r 2^r x^{20-r}$$

Comparing coefficient of x^9 both sides, we get

$$2^{10} \cdot a_9 = 2^{11} a_{11} \Rightarrow \frac{a_9}{a_{11}} = 2$$

(R) Let P be origin and position vectors of point

$$A, B, C \text{ be } \vec{\alpha}, \vec{\beta}, \vec{\gamma}$$

$$|\vec{\alpha} + \vec{\beta} + \vec{\gamma}|^2 + |\vec{\alpha} - \vec{\beta}|^2 + |\vec{\beta} - \vec{\gamma}|^2 + |\vec{\gamma} - \vec{\alpha}|^2 = 3 \left(|\vec{\alpha}|^2 + |\vec{\beta}|^2 + |\vec{\gamma}|^2 \right)$$

$$\Rightarrow 9PG^2 + AB^2 + BC^2 + CA^2$$

$$= 3(\text{PA}^2 + \text{PB}^2 + \text{PC}^2)$$

$$\Rightarrow 9 \times 1 + a^2 + b^2 + c^2 = 3(1 + 4 + 1)$$

$$\Rightarrow a^2 + b^2 + c^2 = 9$$

$$(S) \quad f(xy) = f(x) + f(y)$$

$$\Rightarrow f(16) = f(4) + f(4) = 3 \Rightarrow f(4) = \frac{3}{2}$$

$$\Rightarrow f(2) + f(2) = \frac{3}{2} \Rightarrow f(2) = \frac{3}{4}$$

$$\frac{1}{f(2)} + \frac{1}{f(4)} = \frac{4}{3} + \frac{2}{3} = 2$$

9. Ans (D)

(P) The equation of tangents to hyperbola having slope m are

$$y = mx \pm \sqrt{9m^2 - 49}$$

$$\text{Now, } \frac{2\sqrt{9m^2 - 49}}{\sqrt{1 + m^2}} = 2 \text{ (Given)}$$

$$\Rightarrow m = \pm \frac{5}{2} \Rightarrow \frac{2}{5}|m| = 1$$

(Q) As, $\frac{e}{2}$ and $\frac{e'}{2}$ are eccentricities of a

hyperbola and its conjugate hyperbola, so

$$\frac{4}{e^2} + \frac{4}{e'^2} = 1 \Rightarrow 4 = \frac{e^2 e'^2}{e^2 + e'^2} \dots (1)$$

The equation of the lines $e'x + ey - ee' = 0$

It is tangent to the circle $x^2 + y^2 = r^2$

$$\therefore \frac{ee'}{\sqrt{e^2 + e'^2}} = r = 2 \quad [\text{using (1)}]$$

(R) The equation of common tangents is

$$x - 2y + 4 = 0$$

$$(a + b) = 5$$

(S) The equation of chord to ellipse whose

midpoint is $(0, 3)$ is $(T = S_1)$

$$\frac{x^2}{16} + \frac{y^2}{25} = 1 \Rightarrow \frac{x^2}{16} = 1 - \frac{9}{25} \Rightarrow x = \pm \frac{16}{5}$$

$$\text{Length of chord} = \frac{32}{5} \Rightarrow p = 4$$

10. Ans (A)

$$(P) f'(x) = 0 \Rightarrow 2x \sin \frac{\pi}{x} +$$

$$x^2 \cos \frac{\pi}{x} \left(-\frac{\pi}{x^2} \right) = 0 \Rightarrow \tan \frac{\pi}{x} = \frac{\pi}{2x}$$

$$\Rightarrow \text{Put } \frac{\pi}{x} = \theta \therefore \theta \in (\pi, 21\pi) \text{ as } x \in \left(\frac{1}{21}, 1 \right)$$

Now, number of solution of

$$\tan \theta = \frac{\theta}{2} \text{ in } (\pi, 21\pi) \text{ is } 20$$

(Q) Let a and b roots of $f(x) = 0$

Now, $f(f(1)) = f(f(-2)) = 0$, but

$$f(1) \neq f(-2) \Rightarrow f(1) = a \text{ and } f(-2) = b$$

$$\Rightarrow 1 + p + q + 3 = a, 4 - 2p + q + 3 = b$$

$\Rightarrow$ Adding above two equation

$$2q - p + 11 = a + b \Rightarrow q = -\frac{11}{2}$$

$$\text{Also, } 3p - 3 = a - b \Rightarrow 9p^2 - 18p + 9$$

$$= p^2 - 4(q + 3) \Rightarrow 8p^2 - 18p - 1 = 0$$

$$\Rightarrow g(2) = 4 + 2(8p^2 - 18p) - q + 2 = \frac{27}{2}$$

$$(R) f(x) = (x^9 - 9x^8) + (6x^8 - 54x^7)$$

$$+ (x^7 - 9x^6) + (x^6 - 9x^5) + (x^5 - 9x^4)$$

$$+ (x^4 - 9x^3) + (x^3 - 9x^2) + (x^2 - 9x) + x + 9$$

$$\Rightarrow f(9) = 18$$

$$(S) L = e^{\lim_{x \rightarrow 0} \left(\tan^{-1} \frac{1}{x^2} + x^2 \sin \frac{1}{x^4} \right)} = e^{\frac{\pi}{2}}$$

11. Ans (B)

$$(P) f(z) = \frac{16 \times 16}{2 \times 16} = 8, f(4) = \frac{8 \times 8}{2 \times 8} = 4$$

$$(Q) f(x) = 2x + 2$$

$$(R) f(x) = x^2 + xe^x$$

$$(S) (f(y) - f(x))y^y = x^x f\left(\frac{y^y}{x^x}\right) \text{ differentiate}$$

w.r.t. x keeping y constant

$$-f'(x) \cdot y^y = x^x f\left(\frac{y^y}{x^x}\right) (1 + \ln x) - x^x f'\left(\frac{y^y}{x^x}\right) \frac{y^y (1 + \ln x)}{x^x}$$

$$f'(1) = 1, \text{ put } y^y = x^x, \text{ we get } -f'(x)x^x$$

$$= x^x f(1) (1 + \ln x) - x^x f'(1) (1 + \ln x)$$

$$f(1) = 0, \text{ hence } f'(x) = 1 + \ln x \Rightarrow f(x) = x \ln x$$

$\Rightarrow$ Area bounded between $y = x \ln x$ and

$$y = 2x - 2x^2 \text{ is } \frac{7}{12}$$

12. Ans (A)

$$(P) A^{-1} + B^{-1} = (A + B)^{-1} \Rightarrow (A^{-1} + B^{-1})(A + B) = I$$

$$\Rightarrow I + A^{-1}B + B^{-1}A = 0 \Rightarrow P = A^{-1}B$$

$$I + P + P^1 = 0 \Rightarrow P + P^2 + I = 0$$

$$\Rightarrow P^2 + P^3 + P = 0$$

$$\Rightarrow P^3 = I \Rightarrow |P| = 1$$

$$\Rightarrow |A^{-1}B| = 1 \Rightarrow |A| = |B| = 3$$

(Q)

$$B^2 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

$$(R) \text{ Use } A^{20} = (I + x)^{20}$$

$$(S) a_{22}b_{18} + a_{22}b_{28} + \dots + a_{28}b_{58} =$$

$$\frac{2.8}{1.4} + \frac{2.8}{4.7} + \dots + \frac{2.8}{13.16} = \frac{2.8}{3} \left[1 - \frac{1}{4} + \frac{1}{4} - \frac{1}{7} + \dots + \frac{1}{3} - \frac{1}{16} \right]$$

$$= \frac{16}{3} \times \frac{15}{16} = 5$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (2341)

(P) Using P.I.E. concept required no. of functions

$$= {}^6P_5 - {}^5C_1 \times {}^5P_4 + {}^5C_2 {}^4P_3 - {}^5C_3 {}^3P_2 +$$

$${}^5C_4 {}^2P_1 - {}^5C_5 \times 1$$

$$= 309$$

(Q) Number of non-decreasing

$$= {}^{11+5-1}C_5 = {}^{15}C_5 = 3003$$

$$(R) {}^8C_5 = \frac{8 \times 7 \times 6}{6} = 56$$

$$(S) {}^{10+5-1}C_5 = {}^{14}C_5 = 2002$$

2. Ans (2132)

$$(P) S = \frac{1}{n} \sum_{0 \leq i < j \leq n} (n-i+n-j)^n {}^nC_{n-i} {}^nC_{n-j}$$

$$= 2 \sum_{0 \leq i < j \leq n} {}^nC_i {}^nC_j - S$$

$$\Rightarrow S = \sum_{0 \leq i < j \leq n} {}^nC_i {}^nC_j = \frac{1}{2} \left[\left(\sum_{i=0}^n {}^nC_i \right)^2 - \sum_{i=0}^n {}^nC_i^2 \right]$$

$$= \frac{1}{2} (2^{2n} - 2^n {}^nC_n)$$

$$(Q) \left(\sum_{0 \leq i < j \leq n} ({}^nC_i)^2 + ({}^nC_j)^2 \right) = n \cdot ({}^{2n}C_n)$$

$$(R) S = \sum_{0 \leq i < j \leq n} (n-i+n-j) ({}^nC_{n-i} - {}^nC_{n-j})^2$$

$$2S = 2n \sum_{0 \leq i < j \leq n} ({}^nC_i - {}^nC_j)^2 = 2n \left[n \sum_{i=0}^n {}^nC_i^2 - 2 \sum_{0 \leq i < j \leq n} {}^nC_i {}^nC_j \right]$$

$$S = n \left[(n+1) \sum_{i=0}^n {}^nC_i^2 - ({}^nC_0 + {}^nC_1 + {}^nC_2 + \dots + {}^nC_n)^2 \right]$$

$$= [(n+1) {}^{2n}C_n - 2^{2n}]n$$

$$(S) \sum_{0 \leq i < j \leq n} {}^nC_i {}^nC_j = \frac{1}{2} (4^n - 2^n {}^nC_n)$$

3. Ans (4213)

$$(P) P = \frac{{}^2C_1 \times {}^{14}C_3}{{}^{16}C_4} = \frac{2}{5}$$

$$(Q) P = \frac{{}^{14}C_2}{{}^{15}C_2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{20}$$

$$(R) P = \frac{1}{{}^{15}C_1} \times \frac{1}{2} = \frac{1}{30}$$

$$(S) P = \frac{{}^{14}C_6}{{}^{15}C_7} \times \frac{1}{2} = \frac{7}{30}$$

4. Ans (1143)

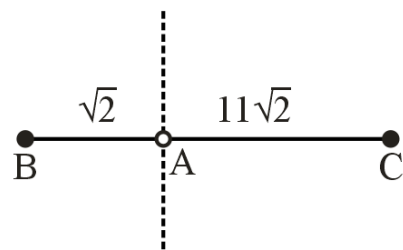
(P) A(5,0) B(7,0)

$$2|z-7| + 6(|z-7| + |z-5|) \geq 0 + 6 \times 2 = 12$$

(Q) A(-4, 0), B(-5, 1), C(7, -11), P(z)

$$AB = \sqrt{2}, BC = 12\sqrt{2}, CA = 11\sqrt{2}$$

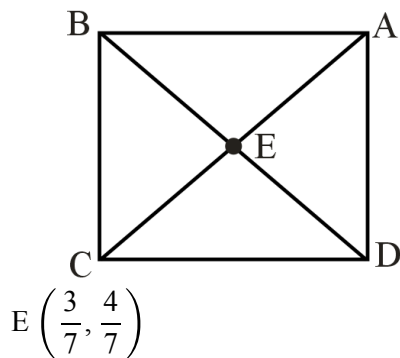
$$AB + AC = BC$$



$$PB + PA + PC \geq BA + AC = 12\sqrt{2}$$

$$\Rightarrow \frac{PA + PB + PC}{\sqrt{2}} \geq 12$$

(R) A(3, 4), B(0, 1), C(0, 0), D(1, 0), P(z)



$$AP + BP + CP + DP \geq AE + EC + BE + ED =$$

$$5 + \sqrt{2}$$

(S) The least value is from centroid of the triangle.

5. Ans (3124)

Given $\frac{atx}{6} = 32$ to here A = (9,0,0), B = (0,6,0)
C = (0,0,C)

(P) Centroid of tetrahedron $(\alpha, \beta, \gamma) \equiv \left(\frac{a}{4}, \frac{b}{4}, \frac{c}{4}\right)$

$$\Rightarrow a = 45\alpha, b = 4\beta, c = 48$$

$$\text{so, } \frac{64\alpha\beta 8}{6} = 32$$

$$xyz = 3$$

(Q) Equation distant point (α, β, γ)

$$= \left(\frac{a}{2}, \frac{b}{2}, \frac{c}{2}\right)$$

$$\Rightarrow a = 2\alpha, b = 2\beta, c = 2\gamma$$

$$\therefore \frac{8\alpha\beta\gamma}{6} = 32$$

$$\therefore xyz = 24$$

(R) equation of plane is $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$

$$\therefore \text{foot of } \perp_r \text{ from origin } \equiv (\alpha, \beta, \gamma) \equiv$$

$$\left(\frac{\frac{1}{a}}{\sum \frac{1}{a^2}}, \frac{\frac{1}{b}}{\sum \frac{1}{a^2}}, \frac{\frac{1}{c}}{\sum \frac{1}{a^2}}\right)$$

$$\text{Let } \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} = t$$

$$\text{So, } \frac{1}{a\alpha} = \frac{1}{b\beta} = \frac{1}{c\gamma} = t$$

$$\text{So, } t = \frac{1}{\alpha^2 + \beta^2 + \gamma^2}$$

$$A = a = \frac{\alpha^2 + \beta^2 + \gamma^2}{\alpha}, b = \frac{\alpha^2 + \beta^2 + \gamma^2}{\beta}, c = \frac{\alpha^2 + \beta^2 + \gamma^2}{\gamma}$$

$$\text{Now } abc = 6 \times 3^2 \text{ or } (\alpha^2 + \beta^2 + \gamma^2)^3 = 192\alpha\beta\gamma$$

(S) Let P be (α, β, γ) then $PA \perp PB$

$$\Rightarrow (\alpha - a) + b(\beta - b) + \gamma\gamma = 0$$

$$\Rightarrow a\alpha + b\beta = \alpha^2 + \beta^2 + \gamma^2$$

$$PB \perp PC \Rightarrow \alpha, \alpha + \beta(\beta - b) + \gamma(\gamma - c) = 0$$

$$\Rightarrow b\beta + c\gamma = \alpha^2 + \beta^2 + \gamma^2$$

$$\therefore \frac{a}{a/\alpha} = \frac{b}{1/\beta} = \frac{c}{1/\gamma} \Rightarrow a = \frac{\alpha^2 + \beta^2 + \gamma^2}{2\alpha},$$

$$b = \frac{\alpha^2 + \beta^2 + \gamma^2}{2\beta}, c = \frac{\alpha^2 + \beta^2 + \gamma^2}{2\gamma}$$

$$\therefore \text{arc} = 6 \times 3^2$$

$$\Rightarrow (\alpha^2 + \beta^2 + \gamma^2)^3 = 192 \times 8\alpha\beta\gamma = 1536\alpha\beta\gamma$$

6. Ans (2431)

$\alpha_1, \alpha_2, \alpha_3, \alpha_4, \& \alpha_5$ are roots of this

equation $x^5 + x^2 + 1 = 0$

put $y = x^2 - 2$

$x^2 = y + 2$

Equation becomes

$$(y + 2)^2 x + y + 3 = 0$$

$$\Rightarrow (y + 2)^4 (y + 2) = (y + 3)^2$$

$$\Rightarrow y^5 + 5y^4 \times 2 + 10 \times y^3 \times 4 + 10y^2 \times 8 +$$

$$5y \times 16 + 32 = y^2 + 6y + 9$$

$$\Rightarrow y^5 + 10y^4 + 40y^3 + 79y^2 + 30y + 23 = 0$$

has roots $\alpha_1^2 - 2, \alpha_2^2 - 2, \dots, \alpha_5^2 - 2$

$$\text{Now } \prod_{i=1}^5 Q(\alpha_i) = -23$$

$$\sum_{i=1}^5 Q(\alpha_i) = -10$$

$$\sum_{1 \leq i < j \leq 5} Q(\alpha_i)Q(\alpha_j) = 40$$

$$\sum_{i=1}^5 (Q(\alpha_i))^2 = 20$$